

# Python String Methods - Practice Tasks

Solutions with code, output and explanation

## Section 1: Case Conversion Methods

### Task 1: Convert "python programming" to uppercase

Code:

```
text = "python programming"
print(text.upper())
```

Output:

```
PYTHON PROGRAMMING
```

**Explanation:** upper() converts every letter of the string to capital letters.

### Task 2: Convert "PYTHON FULL STACK" to lowercase

Code:

```
text = "PYTHON FULL STACK"
print(text.lower())
```

Output:

```
python full stack
```

**Explanation:** lower() converts every letter to small letters.

### Task 3: Convert "python is easy" using capitalize()

Code:

```
text = "python is easy"
print(text.capitalize())
```

Output:

```
Python is easy
```

**Explanation:** capitalize() makes only the first character of the whole string capital and the rest lowercase.

### Task 4: Convert "python full stack developer" using title()

Code:

```
text = "python full stack developer"
print(text.title())
```

Output:

```
Python Full Stack Developer
```

**Explanation:** title() makes the first letter of every word capital.

### Task 5: Convert "PyThOn PrOgRaMmInG" using swapcase()

Code:

```
text = "PyThOn PrOgRaMmInG"
print(text.swapcase())
```

Output:

```
pYtHoN pRoGrAmMiNg
```

**Explanation:** swapcase() changes uppercase letters to lowercase and lowercase letters to uppercase.

## Section 2: Whitespace Removal Methods

### Task 6: Remove spaces from both sides using strip()

Code:

```
text = "  Python  "  
print "[" + text.strip() + "]"
```

Output:

```
[Python]
```

**Explanation:** strip() removes spaces from the beginning and the end. Brackets are printed only to show that the spaces are gone.

### Task 7: Remove spaces only from the left side

Code:

```
text = "  Python  "  
print "[" + text.lstrip() + "]"
```

Output:

```
[Python  ]
```

**Explanation:** lstrip() removes spaces only from the left (start) of the string.

### Task 8: Remove spaces only from the right side

Code:

```
text = "  Python  "  
print "[" + text.rstrip() + "]"
```

Output:

```
[  Python]
```

**Explanation:** rstrip() removes spaces only from the right (end) of the string.

### Task 9: Remove leading and trailing \* characters

Code:

```
text = "****Python****"  
print(text.strip("*"))
```

Output:

```
Python
```

**Explanation:** When a character is given inside strip(), it removes that character from both ends.

### Task 10: Remove leading # characters

Code:

```
text = "####Welcome"  
print(text.lstrip("#"))
```

Output:

```
Welcome
```

**Explanation:** lstrip('#') removes the # characters only from the left side.

## Section 3: Searching Methods

**Task 11: Find the index of 'o' in "Programming"****Code:**

```
text = "Programming"  
print(text.find("o"))
```

**Output:**

2

**Explanation:** find() returns the index of the first match. Indexing starts from 0, so P=0, r=1, o=2.

**Task 12: Find the last occurrence of 'a' in "banana"****Code:**

```
text = "banana"  
print(text.rfind("a"))
```

**Output:**

5

**Explanation:** rfind() searches from the right side and returns the index of the last occurrence.

**Task 13: Find the index of 'g' in "Engineering"****Code:**

```
text = "Engineering"  
print(text.find("g"))
```

**Output:**

2

**Explanation:** find() gives the first position of 'g'. E=0, n=1, g=2.

**Task 14: Find the last occurrence of 'n' in "Programming"****Code:**

```
text = "Programming"  
print(text.rfind("n"))
```

**Output:**

9

**Explanation:** rfind() returns the last position of 'n', which is index 9.

**Task 15: Count the number of 'a' in "banana"****Code:**

```
text = "banana"  
print(text.count("a"))
```

**Output:**

3

**Explanation:** count() returns how many times a character or substring appears in the string.

### **Task 16: Count the number of spaces in "Python Full Stack Developer"**

**Code:**

```
text = "Python Full Stack Developer"
print(text.count(" "))
```

**Output:**

3

**Explanation:** count(' ') counts the space characters. 4 words means 3 spaces.

### **Task 17: Find the first occurrence of "Stack"**

**Code:**

```
text = "Python Full Stack Developer"
print(text.find("Stack"))
```

**Output:**

12

**Explanation:** find() also works with a whole word. It returns the index where "Stack" starts.

### **Task 18: Find the last occurrence of "a" in "abracadabra"**

**Code:**

```
text = "abracadabra"
print(text.rfind("a"))
```

**Output:**

10

**Explanation:** rfind() gives the last index of 'a'. The string has 11 characters, so the last index is 10.

### **Task 19: Count how many times "is" appears**

**Code:**

```
text = "This is Python and this is easy"
print(text.count("is"))
```

**Output:**

4

**Explanation:** count() counts every occurrence of "is", including the ones inside the words "This" and "this".

### **Task 20: Find the index of "Python" in "I love Python Programming"**

**Code:**

```
text = "I love Python Programming"
print(text.find("Python"))
```

**Output:**

7

**Explanation:** find() returns the starting index of "Python". "I love " has 7 characters, so Python starts at 7.

## **Section 4: Replacing Methods**

### **Task 21: Replace "Java" with "Python"**

**Code:**

```
text = "I am learning Java"
print(text.replace("Java", "Python"))
```

**Output:**

```
I am learning Python
```

**Explanation:** replace(old, new) returns a new string where old is replaced by new.

### **Task 22: Replace all spaces with underscores**

**Code:**

```
text = "Python Full Stack Developer"
print(text.replace(" ", "_"))
```

**Output:**

```
Python_Full_Stack_Developer
```

**Explanation:** Every space is replaced with \_ .

### **Task 23: Replace all commas with semicolons**

**Code:**

```
text = "apple,banana,grapes,mango"
print(text.replace(", ", ";"))
```

**Output:**

```
apple;banana;grapes;mango
```

**Explanation:** replace(', '); changes all commas into semicolons.

### **Task 24: Replace "2025" with "2026"**

**Code:**

```
text = "Batch of 2025"
print(text.replace("2025", "2026"))
```

**Output:**

```
Batch of 2026
```

**Explanation:** replace() works with numbers inside a string as well.

### **Task 25: Replace every "a" with "@"**

**Code:**

```
text = "banana"
print(text.replace("a", "@"))
```

**Output:**

```
b@n@n@
```

**Explanation:** replace() replaces all matches by default, so every 'a' becomes '@'.

## **Section 5: Splitting & Joining Methods**

### ***Task 26: Split the sentence into words***

**Code:**

```
text = "Python Full Stack Developer"
print(text.split())
```

**Output:**

```
['Python', 'Full', 'Stack', 'Developer']
```

**Explanation:** split() with no argument splits at spaces and returns a list of words.

### ***Task 27: Split using commas***

**Code:**

```
text = "Apple,Banana,Grapes,Mango"
print(text.split(","))
```

**Output:**

```
['Apple', 'Banana', 'Grapes', 'Mango']
```

**Explanation:** split(',') breaks the string at every comma.

### ***Task 28: Split using -***

**Code:**

```
text = "10-20-30-40"
print(text.split("-"))
```

**Output:**

```
['10', '20', '30', '40']
```

**Explanation:** split('-') breaks the string at every hyphen. The results are still strings, not numbers.

### ***Task 29: Join the list using spaces***

**Code:**

```
words = ["Python", "is", "easy"]
print(" ".join(words))
```

**Output:**

```
Python is easy
```

**Explanation:** join() combines list items into one string, with the given separator between them.

### ***Task 30: Join the list using commas***

**Code:**

```
words = ["HTML", "CSS", "JavaScript"]
print(",".join(words))
```

**Output:**

```
HTML,CSS,JavaScript
```

**Explanation:** Here the separator is a comma, so the items are joined with commas.

### **Task 31: Join the characters of "Python" using -**

**Code:**

```
text = "Python"
print("-".join(text))
```

**Output:**

```
P-y-t-h-o-n
```

**Explanation:** A string is a sequence of characters, so join() puts '-' between each character.

### **Task 32: Split a sentence into words and count the words**

**Code:**

```
sentence = "Python is a very easy language"
words = sentence.split()
print(words)
print("Total words:", len(words))
```

**Output:**

```
['Python', 'is', 'a', 'very', 'easy', 'language']
Total words: 6
```

**Explanation:** split() gives a list of words and len() gives the number of items in that list.

### **Task 33: Split a sentence and join it back using underscores**

**Code:**

```
sentence = "Python is a very easy language"
words = sentence.split()
print("_".join(words))
```

**Output:**

```
Python_is_a_very_easy_language
```

**Explanation:** First split the sentence into a list, then join the list with underscores.

## **Section 6: Boolean Methods**

### **Task 34: Only alphabets in "Python"?**

**Code:**

```
print("Python".isalpha())
```

**Output:**

```
True
```

**Explanation:** isalpha() is True only if all characters are letters.

### **Task 35: Only digits in "123456"?**

**Code:**

```
print("123456".isdigit())
```

**Output:**

```
True
```

**Explanation:** isdigit() is True only if all characters are digits.

**Task 36: Only letters and digits in "Python123"?**

**Code:**

```
print("Python123".isalnum())
```

**Output:**

```
True
```

**Explanation:** isalnum() is True if the string has only letters and digits (no spaces or symbols).

**Task 37: Is "python" completely lowercase?**

**Code:**

```
print("python".islower())
```

**Output:**

```
True
```

**Explanation:** islower() is True when all letters are lowercase.

**Task 38: Is "PYTHON" completely uppercase?**

**Code:**

```
print("PYTHON".isupper())
```

**Output:**

```
True
```

**Explanation:** isupper() is True when all letters are uppercase.

**Task 39: Is "Python123!" alphanumeric?**

**Code:**

```
print("Python123!".isalnum())
```

**Output:**

```
False
```

**Explanation:** It is False because '!' is a special symbol, which is not a letter or digit.

**Task 40: Only digits in "2026"?**

**Code:**

```
print("2026".isdigit())
```

**Output:**

```
True
```

**Explanation:** All characters are digits, so the result is True.

**Task 41: Only alphabets in "python programming"?**

**Code:**

```
print("python programming".isalpha())
```

**Output:**

```
False
```

**Explanation:** It is False because the space between the words is not an alphabet character.

#### **Task 42: Is "HELLO" lowercase?**

**Code:**

```
print("HELLO".islower())
```

**Output:**

```
False
```

**Explanation:** It is False because all the letters are uppercase.

#### **Task 43: Only alphabets in "Hello123"?**

**Code:**

```
print("Hello123".isalpha())
```

**Output:**

```
False
```

**Explanation:** It is False because the string contains digits.

## **Section 7: Prefix & Suffix Methods**

#### **Task 44: Does "https://google.com" start with "https"?**

**Code:**

```
url = "https://google.com"  
print(url.startswith("https"))
```

**Output:**

```
True
```

**Explanation:** startswith() checks the beginning of the string and returns True or False.

#### **Task 45: Does "python.py" end with ".py"?**

**Code:**

```
file = "python.py"  
print(file.endswith(".py"))
```

**Output:**

```
True
```

**Explanation:** endswith() checks the end of the string.

#### **Task 46: Does "student@gmail.com" end with ".com"?**

**Code:**

```
email = "student@gmail.com"  
print(email.endswith(".com"))
```

**Output:**

```
True
```

**Explanation:** The email ends with .com, so the result is True.

### **Task 47: Does "Programming" start with "Pro"?**

**Code:**

```
word = "Programming"  
print(word.startswith("Pro"))
```

**Output:**

True

**Explanation:** The word begins with "Pro", so the result is True.

### **Task 48: Does "notes.pdf" end with ".txt"?**

**Code:**

```
file = "notes.pdf"  
print(file.endswith(".txt"))
```

**Output:**

False

**Explanation:** The file ends with .pdf, not .txt, so the result is False.

## **Section 8: Formatting Method**

### **Task 49: Print "My name is Rahul and I am 22 years old." using format()**

**Code:**

```
name = "Rahul"  
age = 22  
print("My name is {} and I am {} years old.".format(name, age))
```

**Output:**

My name is Rahul and I am 22 years old.

**Explanation:** {} is a placeholder. format() fills the placeholders with the variable values in order.

### **Task 50: Print "The price of Laptop is 55000." using placeholders**

**Code:**

```
item = "Laptop"  
price = 55000  
print("The price of {} is {}".format(item, price))
```

**Output:**

The price of Laptop is 55000.

**Explanation:** The two {} placeholders are replaced by item and price.

## **Bonus Challenge Tasks**

### Challenge 1: Uppercase, replace spaces with \_, count words

#### Code:

```
sentence = input("Enter a sentence: ")
upper = sentence.upper()
print(upper.replace(" ", "_"))
print("Number of words:", len(sentence.split()))
```

Sample input given: python full stack developer

#### Output:

```
Enter a sentence: python full stack developer
PYTHON_FULL_STACK_DEVELOPER
Number of words: 4
```

**Explanation:** upper() makes the text capital, replace() changes spaces to underscores, and split() with len() counts the words. Words are counted from the original sentence.

### Challenge 2: Starts with "https" and ends with ".com"

#### Code:

```
url = input("Enter a URL: ")
print(url.startswith("https") and url.endswith(".com"))
```

Sample input given: https://google.com

#### Output:

```
Enter a URL: https://google.com
True
```

**Explanation:** Both conditions are joined with 'and', so the result is True only when both are True.

### Challenge 3: Names to title case, joined with " | "

#### Code:

```
names = input("Enter names separated by commas: ")
name_list = names.split(",")
result = []
for name in name_list:
    result.append(name.strip().title())
print(" | ".join(result))
```

Sample input given: rahul,priya sharma,amit

#### Output:

```
Enter names separated by commas: rahul,priya sharma,amit
Rahul | Priya Sharma | Amit
```

**Explanation:** The string is split at commas. A loop converts each name to title case (strip() removes extra spaces), and join() combines them with ' | '.

#### **Challenge 4: Count vowels using count()**

**Code:**

```
text = input("Enter a string: ").lower()
vowels = "aeiou"
total = 0
for v in vowels:
    total = total + text.count(v)
print("Number of vowels:", total)
```

*Sample input given: Python Programming*

**Output:**

```
Enter a string: Python Programming
Number of vowels: 4
```

**Explanation:** The text is converted to lowercase first so that capital vowels are also counted. count() is used for each vowel and the counts are added.

#### **Challenge 5: Sentence analysis**

**Code:**

```
sentence = input("Enter a sentence: ")
print("Uppercase:", sentence.upper())
print("Lowercase:", sentence.lower())
print("Title case:", sentence.title())
print("Number of words:", len(sentence.split()))
print("Contains only alphabets:", sentence.isalpha())
print("Starts with a capital letter:", sentence[0].isupper())
```

*Sample input given: Python is easy*

**Output:**

```
Enter a sentence: Python is easy
Uppercase: PYTHON IS EASY
Lowercase: python is easy
Title case: Python Is Easy
Number of words: 3
Contains only alphabets: False
Starts with a capital letter: True
```

**Explanation:** It uses upper(), lower(), title(), split() with len(), isalpha() and isupper() on the first character sentence[0]. isalpha() gives False when the sentence has spaces.